



UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester – Fall 2021

Paper: Advanced Mathematical Economics Course Code: ECON-307-A

Roll No. 37686...

Time: 3 Hrs. Marks: 60

(6x5=30)

Q.1. Answer the following short questions:

- I. Explain the concept of Maximum Principle.
- II. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$
- III. Find characteristic roots of: $(r+4)(r+4)(r^2 + 2r + 2) = 0$
- IV. Write complementary function of characteristic equation $(r+4)(r+4)(r^2 + 2r + 2) = 0$.
- V. Explain the concept of Primal and Dual.
- VI. Explain Two Variable Phase Diagrams.

Answer the following questions.

(3x10=30)

- Q. 2. Test convergence of the path of $y_{t+2} - 8y_{t+1} + 2y_t = 10$, using Schur Theorem.
- Q. 3. Using Routh Theorem, find dynamic stability of $y'''(t) + 4y''(t) + 5y'(t) - 2y = -2$
- Q. 4. Write Kuhn Tucker Conditions for numerical, $C = (x_1 - 4)^2 + (x_2 - 4)^2$, $2x_1 + 3x_2 \geq 6$, $-3x_1 - 2x_2 \geq -12$, $x_1, x_2 \geq 0$. Further solution not required.

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UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program : Fifth Semester - 2021

Paper: Advanced Mathematical Economics

Course Code: ECON-307 A

Part - II

Time: 2 hrs 45 min

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Give short answers of the following:

(5x5=25)

I. Find the roots of the equation $3x^2 - x + 1 = 0$

II. What are exact differential equations?

III. Briefly explain Routh Theorem.

IV. Explain complex numbers

V. Solve $\frac{dy}{dt} + 4y = 2$ where $y(0) = 4$.

Give brief answers of the followings.

(2x5=10)

Q. 3. Find the solution of $y''(t) + 9y'(t) - 10y = 5$, $y(0) = 6$, $y'(0) = 8$

(10)

Q. 4. Explain the concept of linear programming with reference to its application in economics. (10)

Q. 5. Solve the following difference equation and verify your answer:

(10)

 $y_{t+2} + 6y_{t+1} + 9y_t = 4$ where $y_0 = 1$ and $y_1 = 3$.

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B.S. 4 Years Program : Fifth Semester – 2021

Paper: Advanced Mathematical Economics
Course Code: ECON-307-A

(Re-Conduct)

Part – II

Roll No.

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions:

(5x4=20)

I. Kuhn Tucker Conditions

II. Concept of Primal and Dual

III. Two Variable Phase Diagrams

IV. Calculus of Variations

V. The Maximum Principal

Answer the following questions.

(3x10=30)

Q. 3. Find the Particular Solution of:

(5+5)

i. $y'''(t) + 3y''(t) = \frac{1}{3}$

ii. $y'''(t) + y''(t) + 3y'(t) = 1$

Q.4. Apply Routh Theorem and find the dynamic stability of the time-path (without calculating the characteristic roots): $y'''(t) - 2y''(t) - y'(t) + 2y = 4$

(10)

Q. 5. Test the convergence of the path of $y_{t+2} + 3y_{t+1} + 2y_t = 12$, using Schur Theorem.

(10)



THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

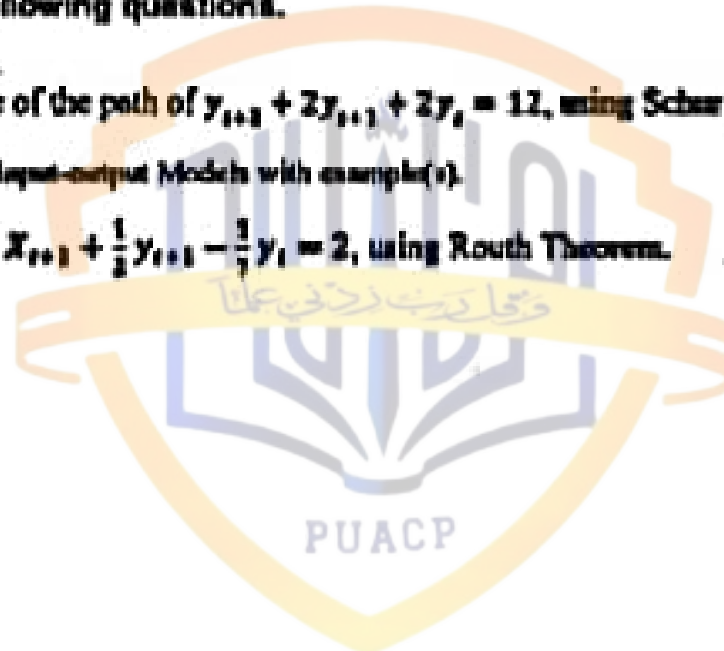
(6x5=30)

- I. Find Particular Solution of: $y^{(4)}(t) + 4y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$
- II. Find characteristic roots of: $(r+2)(r+2)(r^2 + 2r + 2) = 0$
- III. Write complementary function of characteristic equation $(r+2)(r+2)(r^2 + 2r + 2) = 0$.
- IV. Explain Routh Theorem.
- V. Explain the concept of Primal and Dual.
- VI. Explain use of Linear Programming in Economics.

Answer the following questions.

(3x10=30)

- Q. 2. Test convergence of the path of $y_{t+1} + 2y_{t+1} + 2y_t = 12$, using Schur Theorem.
- Q. 3. Explain Dynamic Input-output Models with example(s).
- Q. 4. Solve the model: $x_{t+1} + \frac{1}{2}y_{t+1} - \frac{1}{7}y_t = 2$, using Routh Theorem.





UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Fifth Semester – Fall 2022

Paper: Advanced Mathematical Economics

Course Code: ECON-307-A

Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions.

(6x5=30)

- I. Explain the concept of Calculus of Variations.
- II. Find Particular Solution of: $y^{(4)}(t) + 6y'''(t) + 14y''(t) + 16y'(t) + 8y = 2$
- III. Find characteristic roots of: $(r+3)(r+3)(r^2 + 2r + 2) = 0$
- IV. Write complementary function of characteristic equation $(r+3)(r+3)(r^2 + 2r + 2) = 0$.
- V. Explain the concept of Primal and Dual.
- VI. Explain The Maximum Principal.

Answer the following questions.

(3x10=30)

- Q.2. Test convergence of the path of $y_{t+2} - 4y_{t+1} + y_t = 12$, using Schur Theorem.
- Q.3. Elaborate Inflation-Unemployment Model using Simultaneous Differential and Difference Equations.
- Q.4. Write Kuhn Tucker Conditions for numerical, $U = xy$, $x + y \leq 100$, $x \leq 40$, $x, y \geq 40$. Further solution not required.

